SYMBOLIC POWERS OF A HEIGHT-ONE PRIME IN A COMPLETE LOCAL DOMAIN ARE EVENTUALLY CONTAINED IN THE ORDINARY POWERS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The symbolic power of a prime ideal as the kernel of the map to the localized quotient, and the Mittag-Leffler condition for an inverse system of modules. Introduced in the author's section on Hartshorne-Lichtenbaum vanishing, where the counted statement occurs, and used at the start of the counted proof. Supplies the objects of the statement and the criterion by which the proof controls the inverse system.
- Faithful flatness of the completion of a Noetherian local ring. Used inside the counted proof at source lines 3715-3718. Lets the proof conclude from vanishing after completion that the element in question is already zero.

Lemma 0.1. *Let $(A, \mathfrak{m})$ be a complete Noetherian local domain. Let $\mathfrak{p} \subset A$ be a prime ideal of dimension 1. For every $n \geq 1$ there is an $m \geq n$ such that $\mathfrak{p}^{(m)} \subset \mathfrak{p}^n$.*

Proof. Recall that the symbolic power $\mathfrak{p}^{(m)}$ is defined as the kernel of $A \to A_\mathfrak{p}/\mathfrak{p}^m A_\mathfrak{p}$. Since localization is exact we conclude that in the short exact sequence

$$0 \to \mathfrak{a}_n \to A/\mathfrak{p}^n \to A/\mathfrak{p}^{(n)} \to 0$$

the support of $\mathfrak{a}_n$ is contained in $\{\mathfrak{m}\}$. In particular, the inverse system $(\mathfrak{a}_n)$ is Mittag-Leffler as each $\mathfrak{a}_n$ is an Artinian A-module. We conclude that the lemma is equivalent to the requirement that $\lim \mathfrak{a}_n = 0$. Let $f \in \lim \mathfrak{a}_n$. Then f is an element of $A = \lim A/\mathfrak{p}^n$ (here we use that A is complete) which maps to zero in the completion $A_\mathfrak{p}^\wedge$ of $A_\mathfrak{p}$. Since $A_\mathfrak{p} \to A_\mathfrak{p}^\wedge$ is faithfully flat, we see that f maps to zero in $A_\mathfrak{p}$. Since A is a domain we see that f is zero as desired. □

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